

the jet velocity can be assumed to remain constant. Therefore, the horizontal distance traveled by jet (x) in time (t) is equal to:

$$x = v.t \quad (7)$$

The vertical component of the trajectory of the jet will have a constant acceleration downward due to the force of gravity. Therefore, at any time, t, the y-position of the jet may be calculated as:

$$y = \frac{1}{2}gt^2 \quad (8)$$

Rearranging Equation (8) gives:

$$t = \left(\frac{2y}{g}\right)^{0.5} \quad (9)$$

Substitution of t and v from Equations 9 and 2 into Equation 7 results in:

$$x = C_v\sqrt{2gh}\left(\frac{2y}{g}\right)^{0.5} \quad (10)$$

Equations (10) can be rearranged to find  $C_v$ :

$$C_v = \frac{x}{2\sqrt{yh}} \quad (11)$$

Therefore, for steady flow conditions (i.e., constant h in the head tank), the value of  $C_v$  can be determined from the x, y coordinates of the jet trajectory. A graph of x plotted against  $\sqrt{yh}$  will have a slope of  $2C_v$ .

## 7.2. DETERMINATION OF THE COEFFICIENT OF DISCHARGE

If  $C_d$  is assumed to be constant, then a graph of Q plotted against  $\sqrt{h}$  (Equation 6) will be linear, and the slope of this graph will be:

$$s = C_d A_o \sqrt{2g} \quad (12)$$